1 MATLAB Programming

1.1 Analysis of a characteristic curve

The following problem deals with a simplified characteristic curve of a sensor or the torque response of a drive.

a. Determine the roots, Min/Max values and points of inflection of the polynominal function f(x)= 0 (Implement all Matlab commands inside a script).

$$f(x) \quad = 4x^4 - 2x^3 - 18x^2 + 8x + 8$$

b. Plot the function f(x) between the range -2 and 2
   i. with a step size 0.5 for x.
   ii. with a step size of 0.1 for x.

c. What are the differences between the plot i. and ii. and why are they different?

1.2 Electric kettle (experiment data and simulation)

This exercise is splitted in two parts:



a. Practical experiment data

Task: Boil water (1.0l) from ambient temperature to 95°C (Go not higher than 95°C!) in an electric kettle and measure the temperature and time.

Fill out a table and write down the following parameter: time (stepsize = 5s for the first 40 s, 10s for time > 40s), temperature [°C].

i. Create two vectors in MATLAB: MeasTime,MeasTemp and program in a script a graph which shows the temperature over the time including lables, title etc.

You can find the recorded data on the next page.

Due to the current situation you can find the measured values of the experiment here.

time [s]	temperature [°C]
0	22,80
5	22,80
10	22,80
15	22,80
20	25,70
30	32,10
40	36,90
50	42,70
60	45,80
70	51,70
80	55,40
90	59,10
100	64,40
110	68,90
120	73,10
130	77,80
140	82,10
150	85,90
160	92,80

b. Simulation part

Simulate the experiment in Matlab.
Technical Information:
Power of electric kettle: P=2000 W
Mass of water: m=1 kg (is equal 1 l)
$\Delta\vartheta$: 90°C- start temperature °C

Heat capacity: c= 4,1813 kJ/kg*K

Calculations:

$$\Delta\vartheta = \frac{P * t}{m * c}$$

1 J = 1 Ws

Extend the script for simulate the temperature vs time. The simulation stops at 90 °C. Define a start temperature (same value from the experiment) and increase the time in steps of 1 s.

i. How long does it take for to heat up the water from start temperature to 90°C?

ii. Compare the result (simulation) against the data from experiment (e.g. create a subplot).

iii. What are the different between experiment and simulation? What are the reason for the result?

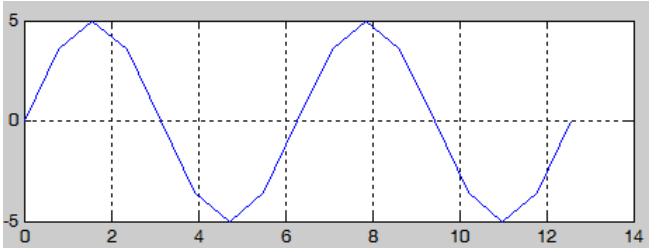
iv. Find out, by using the simulation and results from experiment, of what is more efficiency:

i. Boil 1 l water
ii. Boil 0.5 l water twice

Justify and explain the solution to the tutor!

1.3 Signal reconstruction

The sampling rate of the given sine signal is not high enough.
u = 5V, x = 0 to 4*pi in steps of pi/4. y = u*sin(x)



Create a script and use the Matlab function "polyfit" for reconstruct the signal with a higher accuracy. Show the results in plot(s) and find out the grade of the polynomial which best fits to real data.

1.4 NTC Thermistor Characteristic Curve

Temperature sensors such as NTC (Negative Temperature Coefficient) thermistors are widely used in mechatronic systems for thermal monitoring, but they exhibit a non-linear relationship between resistance and temperature.

Implement the exponential resistance-temperature relationship in a MATLAB script:

$$R(T) = R_0 \cdot \exp\left(B \cdot \left(\frac{1}{T} - \frac{1}{T_0}\right)\right)$$

(Given parameters: Nominal resistance $R_0 = 10\,\text{k}\Omega$ at reference temperature $T_0 = 25°\text{C}$ ($298.15\,\text{K}$), material constant $B = 3950\,\text{K}$).

a) Calculate the resistance values for a temperature range from $0°\text{C}$ to $100°\text{C}$ with a step size of $1°\text{C}$ (make sure to correctly convert Celsius to Kelvin for the calculation).

b) Program a script to plot the resistance over temperature. Include proper axis labels with units, a title, and a grid.

c) Use MATLAB to automatically determine and display at which exact temperature the sensor resistance drops below an alarm threshold of $2\,\text{k}\Omega$.